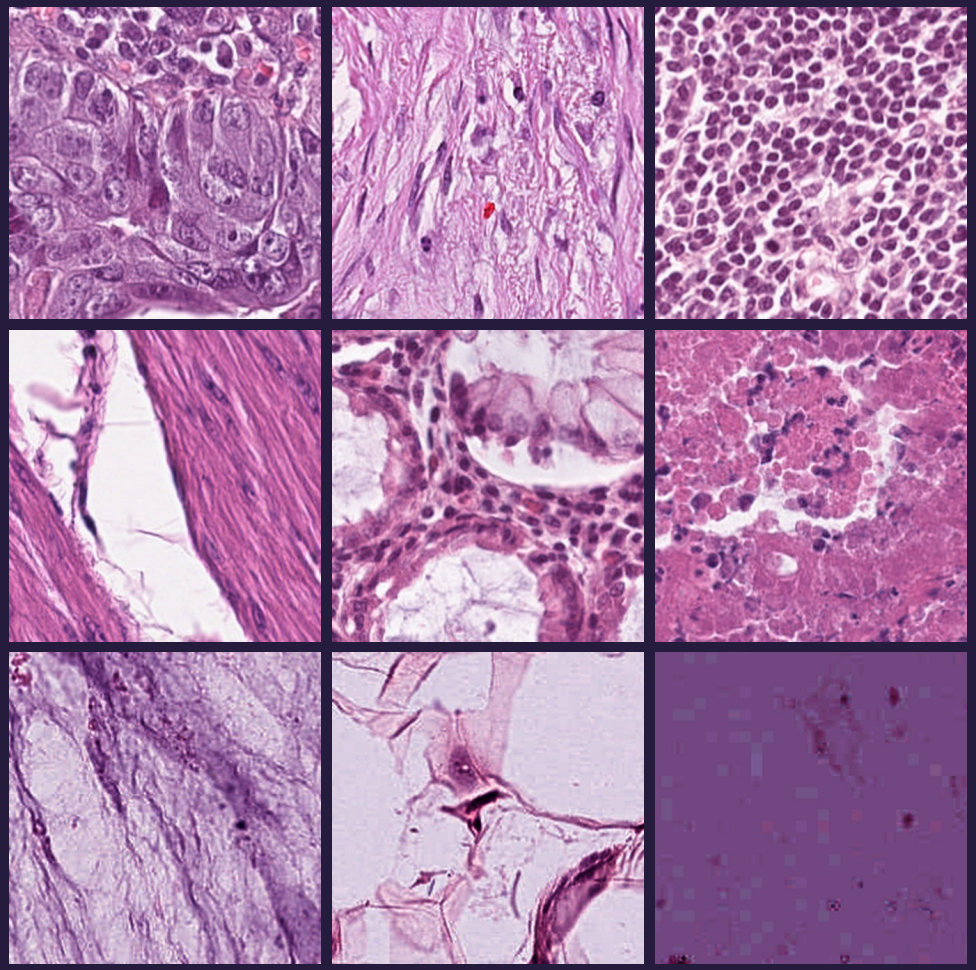
# TwinRead

Two pathology foundation models as two independent readers, and a second look that runs offline



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2026 Ashby Prize in Computational Science Hackathon

Research prototype, not a diagnostic device

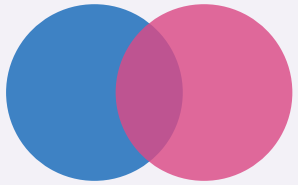
# Pathology reads vary, and second readers are scarce

**75.3%**

of individual breast-biopsy reads matched the expert panel, and only 48% for atypia

**47%**

of the world’s population has little or no access to diagnostics



## Can two AI models act as independent readers?

And can that second look run offline, on a small board?

Elmore JG et al., JAMA 2015 (115 pathologists, 6,900 interpretations). Fleming KA et al., Lancet Commission on diagnostics, Lancet 2021.

# Two models, trained differently, miss different tiles

**CONCH**
Contrastive: 1.17 M image–caption pairs

**MUSK**
Masked: 50 M images and 1 B text tokens

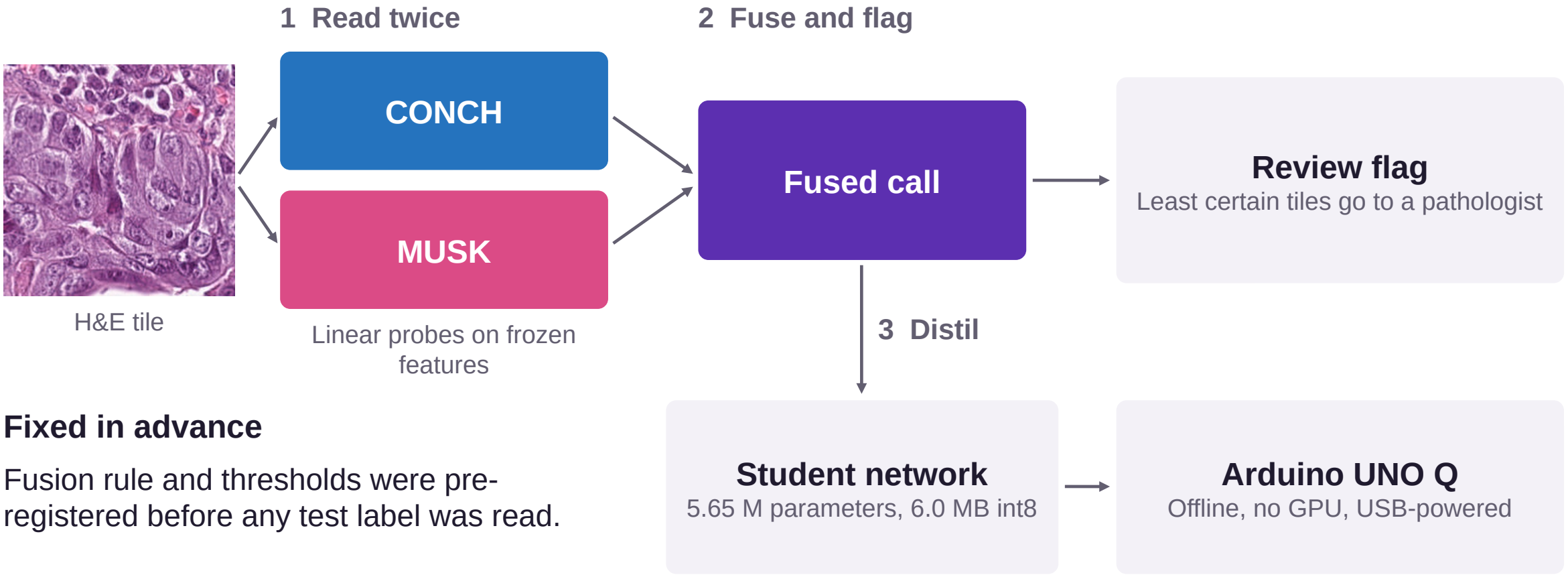
**505 of 7,180 colorectal test tiles have at least one reader wrong**




**Only 110 tiles fool both. The rest are gaps the other reader can cover.**

Linear probes on frozen features, CRC-VAL-HE-7K, 9 tissue classes. Lu et al., Nat Med 2024; Xiang et al., Nature 2025.

# Read twice, flag doubt, then shrink to a small board




## Fixed in advance

Fusion rule and thresholds were pre-registered before any test label was read.

Train: 17,349 NCT-CRC-HE-100K tiles. Test: 7,180 CRC-VAL-HE-7K tiles (9 classes) and 1,500 Camelyon17 tiles from an unseen hospital.

# CUDA kernels: ×202 preprocessing, ×10.6 augmentation

**×202**

## K1: dual-reader preprocessing

One tile in, CONCH 448 px and MUSK 384 px out, bit-exact to the official transforms

**×10.6**

## K2: training augmentation

Nine dependent stages plus JPEG, each within one grey level of a float64 reference







**×1.11** whole embedding run: the model forward is 89% of the time

**×2.49** whole student training step: 1,559 to 3,880 images/s

RTX 5060; compile-checked for Delta A100 and DeltaAI GH200; compute-sanitizer clean. Tiles/s (K1), images/s (K2). K2 simulates a USB camera filming a screen.

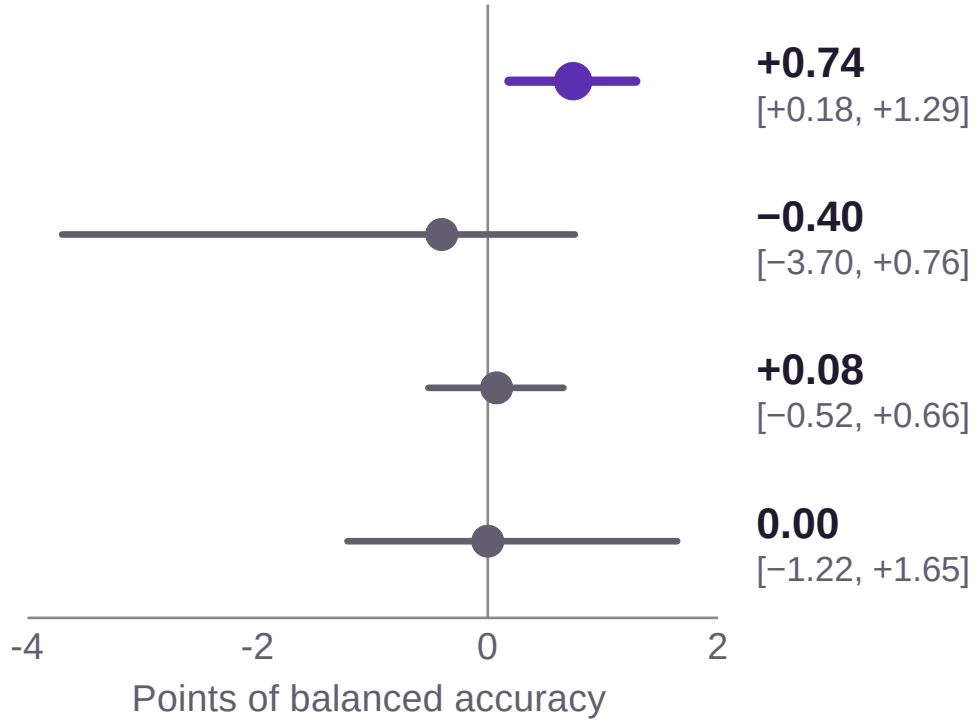
# Fusion adds +0.74 points on colorectal tiles only

**Balanced accuracy, colorectal test set**

- **94.13%** CONCH alone
- **94.80%** MUSK alone
- **95.55%** CONCH + MUSK fused

**+3.6 points of tumour recall: most of the net gain**

**Fused minus best single reader, 95% CI**



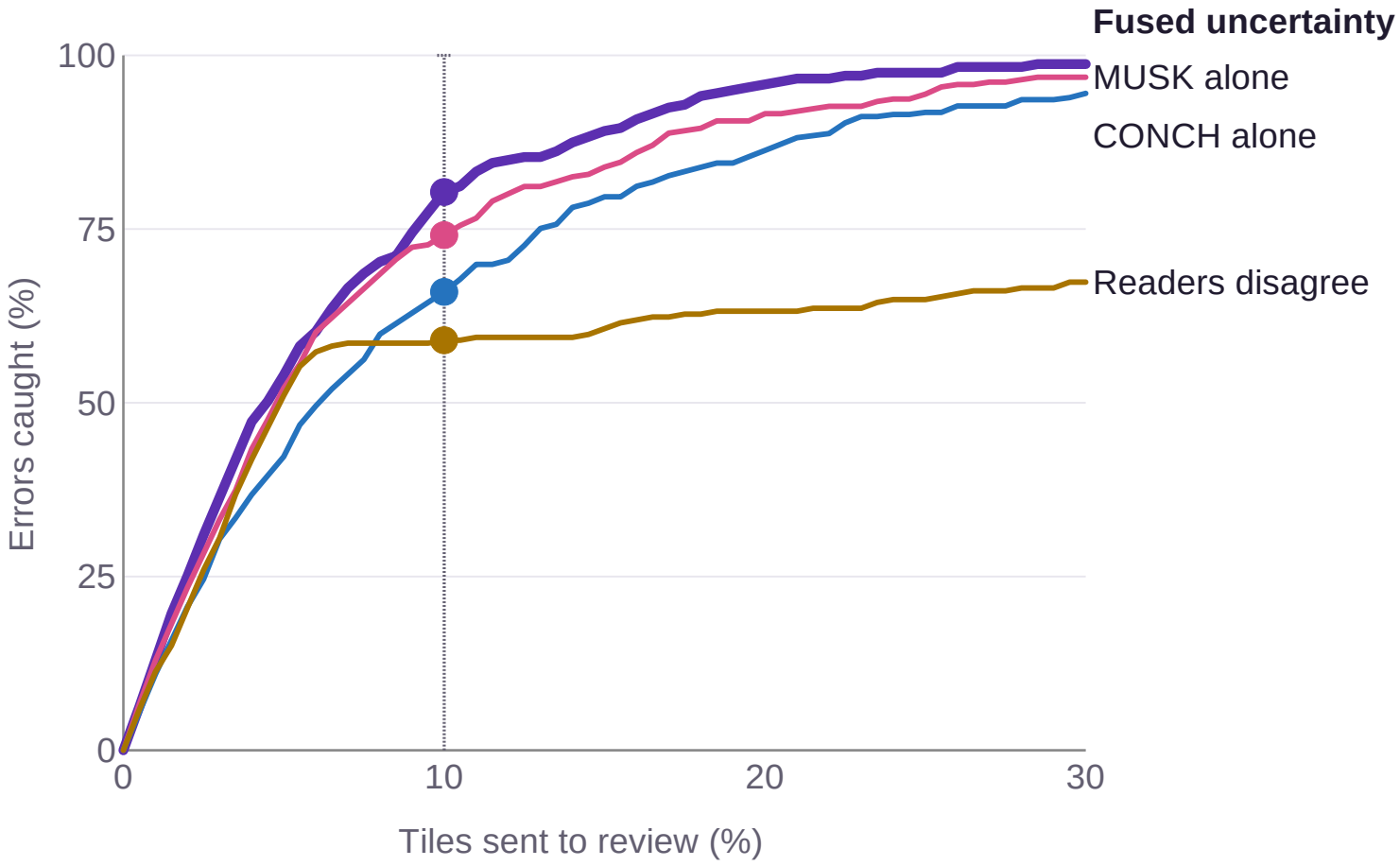

CIs: tile-level on colorectal (anti-conservative, no slide IDs), slide-clustered on Camelyon17. Test sets: 7,180 and 1,500 tiles.

# Reviewing the least certain 10% catches 80% of errors

**Errors caught at a 10% review budget**

- **80.3%**
  **Fused reader’s uncertainty**
- **74.1%** MUSK alone
- **66.0%** CONCH alone
- **59.0%** Readers disagree

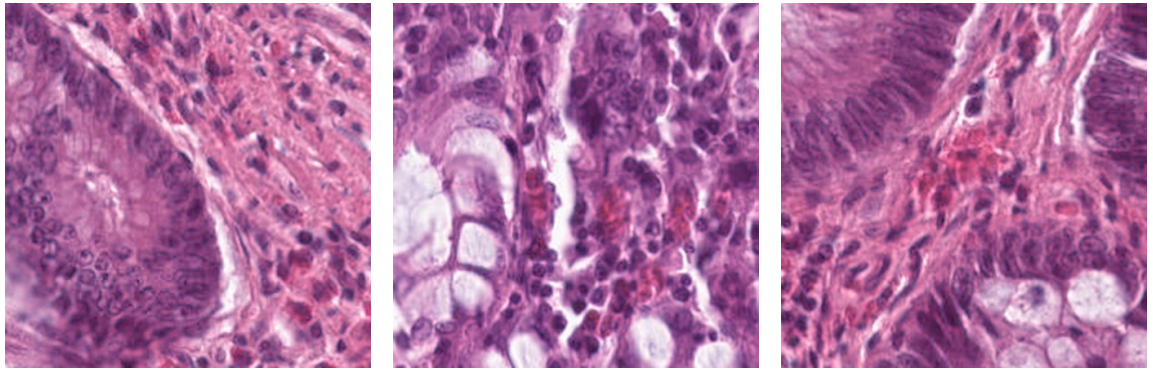
**The signal is the fused reader’s doubt, not disagreement.**



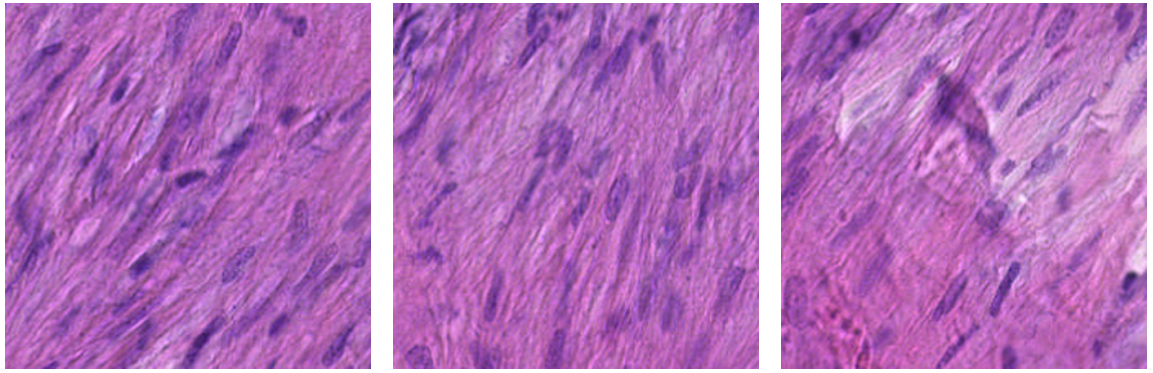

CRC-VAL-HE-7K, 7,180 tiles. Single readers flag their own errors by their own confidence. Disagreement covers only 5.6% of tiles, so the rest of its budget is effectively random.

# 53 tiles fooled both readers; six also escaped the flag

**Truth: tumour. Both readers: normal.**



**Truth: stroma. Both readers: muscle.**



**53** tiles (0.74%): both wrong, same call, both at least 0.80 confident

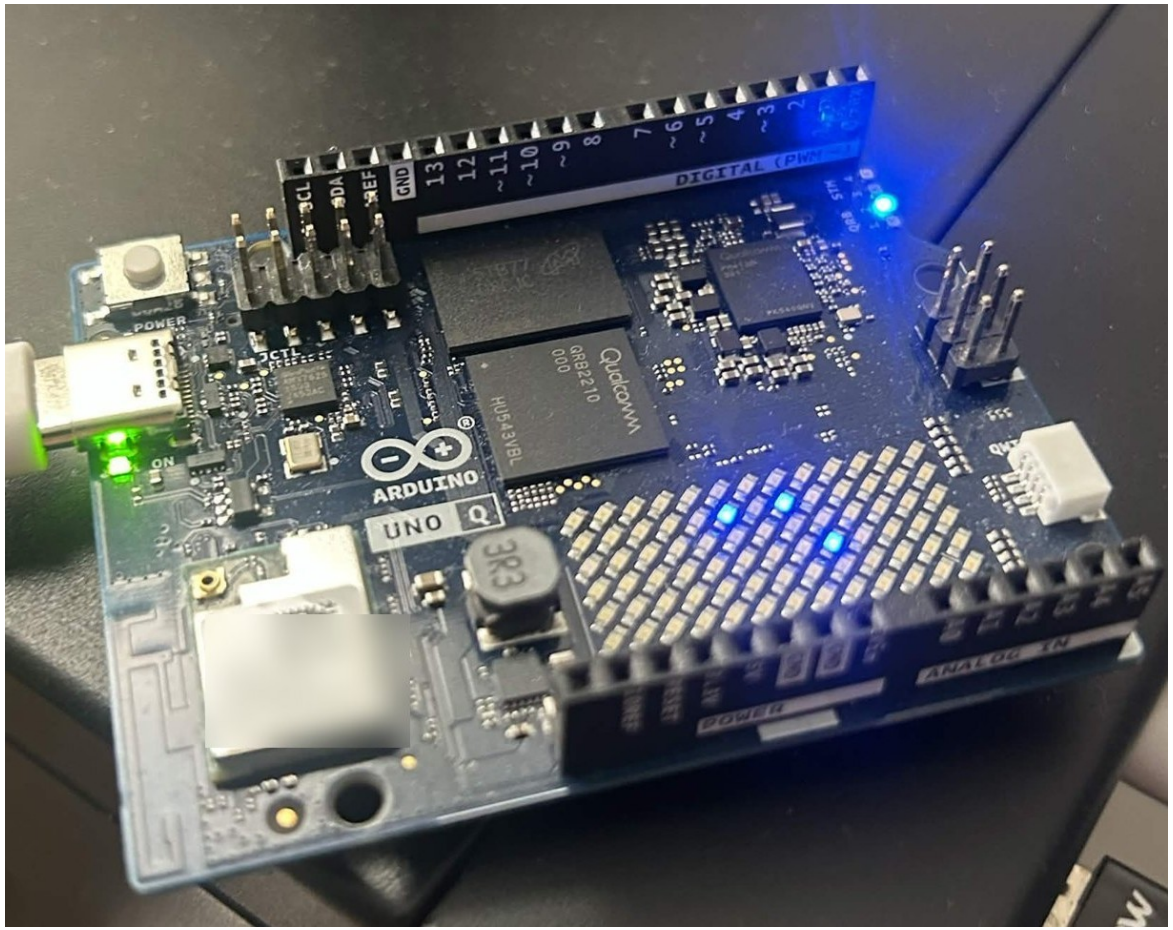
**47** caught by the pre-registered flag, which reviews 25.1% of tiles

**6** escaped it (shown): both readers at least 0.98 confident

**Two models agreeing is not proof. A pathologist stays in the loop.**

CRC-VAL-HE-7K, truth as labelled in the dataset. 36 of the 53 are muscle vs stroma; 11 are tumour called normal.

# A 6 MB student runs offline on a USB-powered board




## 18.7 ms

per tile, model inference: int8, 4 threads, offline

## 93.05%

balanced accuracy on all 7,180 test tiles, below every teacher (94.13–95.55%)

## 6.0 MB

5.65 M parameters: 1.4% of the two teachers' vision towers

All measured on the Arduino UNO Q: Qualcomm QRB2210, 4× Cortex-A53, 3.6 GiB RAM, onnxruntime on CPU. Stored test tiles; no camera captures yet.

# Three gaps remain, each with a concrete next step

| What remains | Next step |
| --- | --- |
| The fusion gain is fragile: validation chose the rule by 1–2 tiles | → **Rerun on all 100k training tiles on NCSA Delta (kit ready)** |
| The pre-registered threshold flagged 25.1% of tiles, not 10% | → **Calibrate per site; add slide IDs for clustered CIs** |
| The board’s flag tracks the teachers weakly (κ 0.32), on stored tiles only | → **Real camera captures on the board, then clinical validation** |

## Overcome along the way

**Non-commercial licences**
open control pair, shareable bundle

**int8 calibration collapse**
percentile calibration

**Low x86 int8 scores**
traced to an onnxruntime kernel

# Why it matters to pathologists and patients

## For pathologists

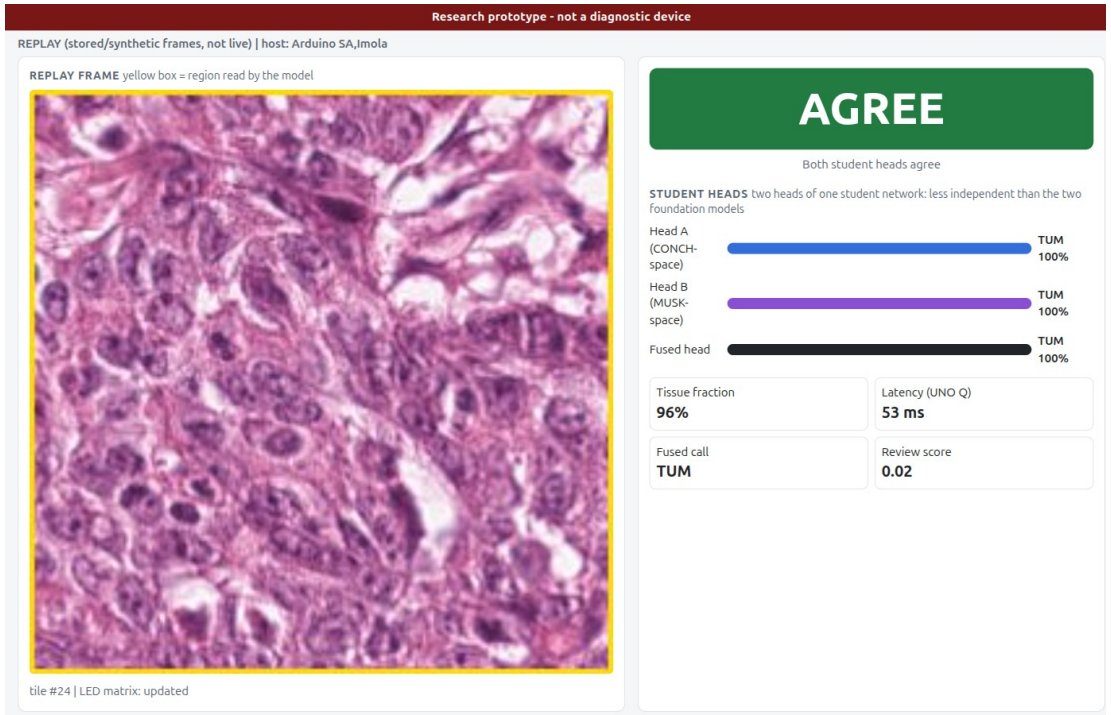
A short list of tiles worth re-checking, with each reader’s call shown.

## For patients

A low-cost offline second look could help where no second pathologist is available.

## For the field

Two readers help by making a better-calibrated reader, not through disagreement.




The app on the UNO Q, replaying a stored test tile. Its two heads share one student network.

# TwinRead in three numbers

**+0.74 pts**

balanced accuracy from fusion on colorectal tiles; no gain on Camelyon17

**80.3%**

of errors caught by reviewing the least certain 10% of tiles

**18.7 ms**

per tile on an offline board: a 6.0 MB model, 93.05% accurate

## Thank you. Questions?

Yubo Wu, Noah Zhang, William Liu, Franklin Liu, Andy Chen

github.com/iujab/twinread-conch-musk-edge

# References


Elmore JG, et al. Diagnostic concordance among pathologists interpreting breast biopsy specimens. JAMA 2015. doi:10.1001/jama.2015.1405

Fleming KA, et al. The Lancet Commission on diagnostics: transforming access to diagnostics. Lancet 2021. doi:10.1016/S0140-6736(21)00673-5

Lu MY, et al. A visual-language foundation model for computational pathology (CONCH). Nat Med 2024. doi:10.1038/s41591-024-02856-4

Xiang J, et al. A vision–language foundation model for precision oncology (MUSK). Nature 2025. doi:10.1038/s41586-024-08378-w

Ikezogwo WO, et al. Quilt-1M: one million image-text pairs for histopathology (QuiltNet). NeurIPS 2023. arXiv:2306.11207

Li et al. Multi-modal foundation models for computational pathology: a survey. TMLR 2025. PMC13081565

Muneer A, et al. Review on multimodal data integration for cancer research. Artif Intell Rev 2026. doi:10.1007/s10462-026-11522-9

Kather JN, et al. NCT-CRC-HE-100K and CRC-VAL-HE-7K (CC-BY-4.0). Bandi P, et al. Camelyon17; Koh PW, et al. WILDS (Camelyon17-WILDS, CC0).

Open models: PathGen-CLIP (CC-BY-4.0), QuiltNet-B-16 (MIT), timm mobilenetv4_conv_small.


Code, results JSON and this deck’s generator: github.com/iujab/twinread-conch-musk-edge. Every TwinRead number here is read from results/*.json at build time.

# Backup: data, splits and statistics

| Split | Data | Tiles | Used for |
|---|---|---|---|
| Train | NCT-CRC-HE-100K subset | 17,349 | Probes and student |
| Validation | Same source, group-disjoint | 2,658 | Fusion rule, thresholds |
| Test | CRC-VAL-HE-7K, 9 classes | 7,180 | Evaluated once, after pre-registration |
| Test | Camelyon17-WILDS, hospital 2 | 1,500 | Unseen hospital; trained on hospitals 0, 3, 4 |

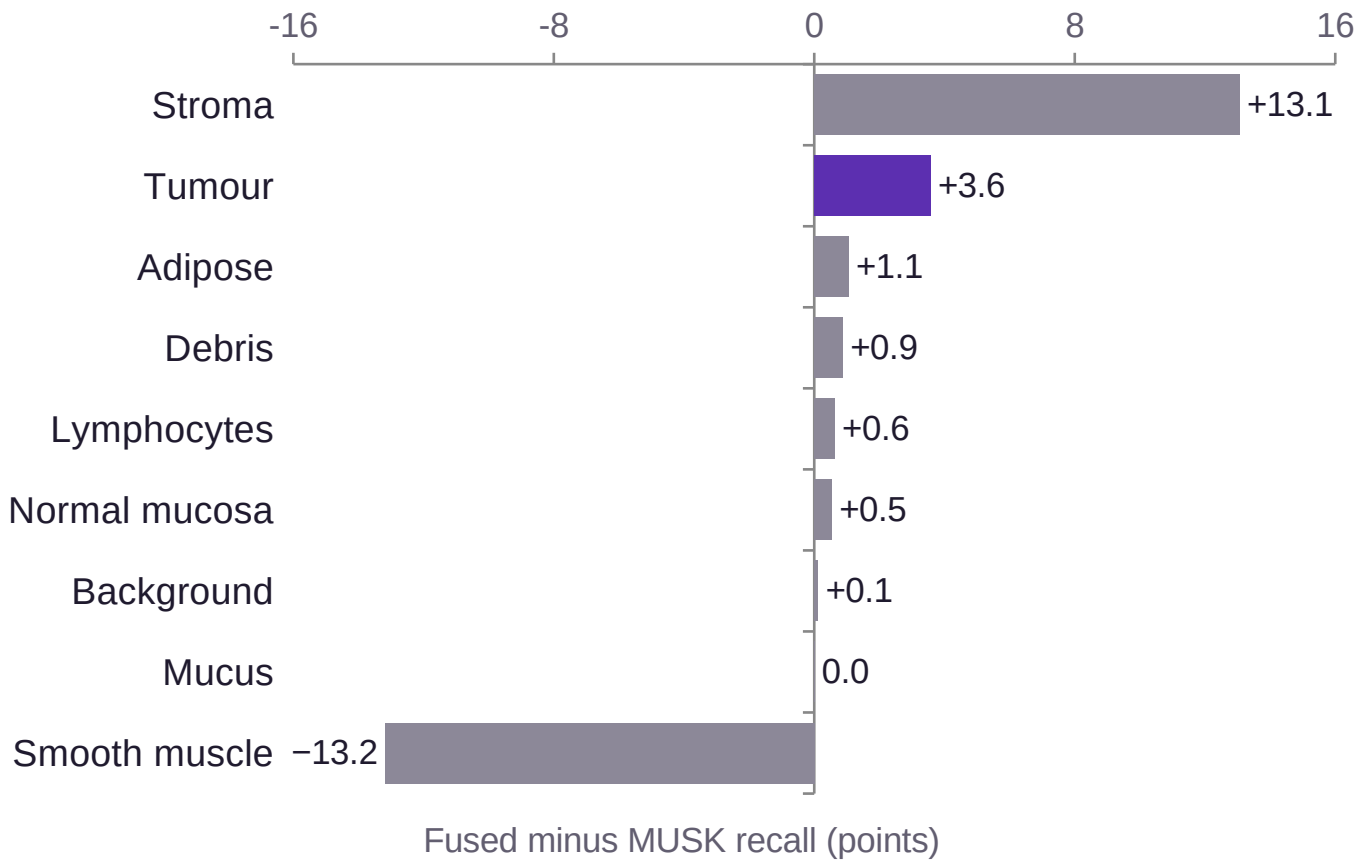
**Statistics.** Percentile bootstrap, 1,000 resamples. Colorectal CIs and McNemar tests are tile-level and anti-conservative because the dataset has no slide IDs; Camelyon17 CIs are clustered by slide.

**Licences.** CONCH and MUSK are CC-BY-NC-ND, so their student is never distributed. PathGen-CLIP + QuiltNet-B-16 is the control pair and the shareable bundle.

**Scale.** Teacher numbers are from a local 20k-tile subset on one RTX 5060. The Delta kit is written and tested locally and has not run on Delta.

Sources: results/cm_local/CM/*/preregistration.json, results/student/cm_crc/metrics.json, ncsa/README.md.

# Backup: why the colorectal gain is fragile




## +0.40 of the +0.74

points of balanced accuracy come from tumour recall. Muscle and stroma cancel.

## 2,641 / 2,640 / 2,639

of 2,658 validation tiles correct for the average, gated MLP and concat rules. Validation is near ceiling.

## −0.40

test points for the runner-up rule, the gated MLP, against MUSK.

Post hoc and descriptive. Balanced accuracy is the mean of 9 class recalls, so each class contributes its recall change divided by 9. Source: results/headline.json.

# Backup: where the review flag falls short

| At a 10% budget | Fused uncertainty | Learned flag | Best single reader | Flag: cleared-tile accuracy | Best single: cleared-tile accuracy |
|---|---|---|---|---|---|
| Colorectal | 80.3% | 80.3% | 74.1% | 99.27% | 98.85% |
| Camelyon17 | 81.1% | 75.5% | 74.5% | 99.04% | 99.11% |

| Pre-registered threshold on test | Target flagged | Actually flagged | Errors caught |
|---|---|---|---|
| Colorectal | 10% | 25.1% | 97.5% |
| Camelyon17 | 10% | 5.3% | 58.5% |

Validation sits near ceiling (99.36%), and validation to test is a distribution shift, so a threshold calibrated on it does not transfer. Recalibrate on the target site.

Single readers flag their own errors by their own confidence. Source: results/cm_local/CM/*/review.json, results/headline.json.

# Backup: kernel throughput

| K1: tiles/s, both readers, 224 px | Rate |
|---|---|
| PIL, 1 CPU process | 221 |
| PIL DataLoader, 8 workers | 538 |
| torch GPU ops (exact) | 1,340 |
| Triton (our baseline) | 82,058 |
| K1 CUDA, including host copy | 46,157 |
| **K1 CUDA, input on GPU** | **90,131** |

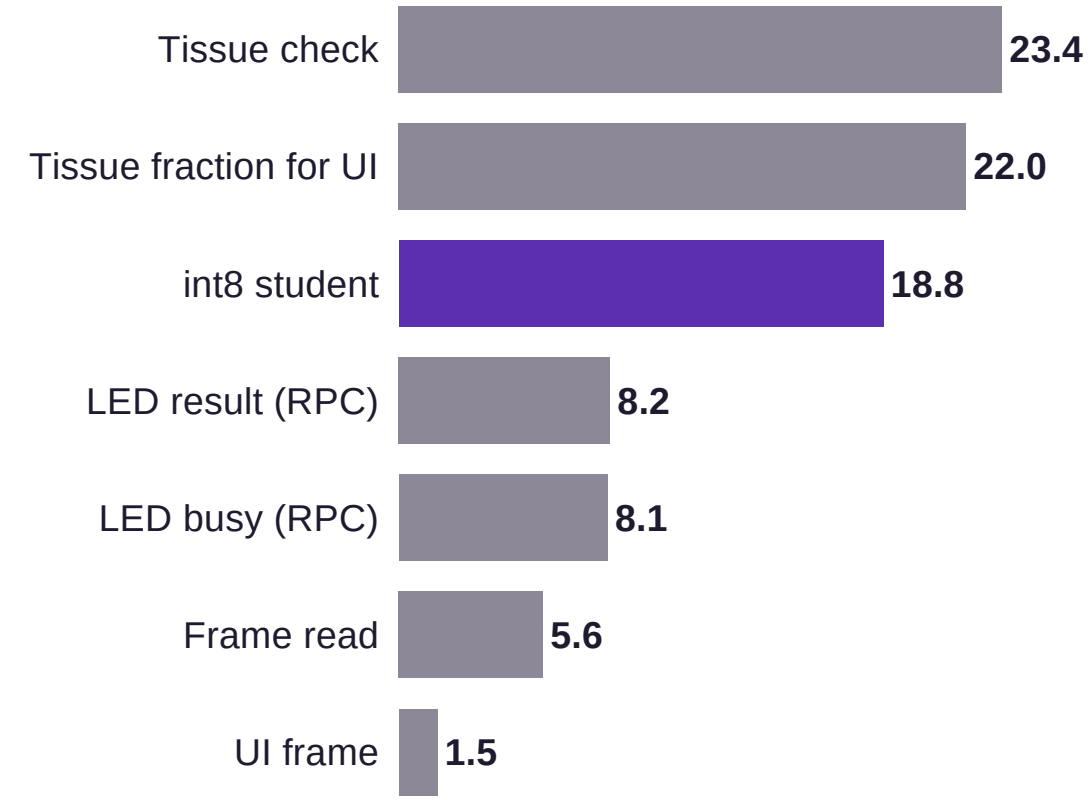
| K2: images/s, severity 1.0 | Rate |
|---|---|
| numpy, 1 process | 44 |
| numpy, 4 processes | 166 |
| torch GPU ops | 1,097 |
| **K2 CUDA** | **46,315** |

K1 cosine to the official transforms: at least 0.999997. The ×1.11 end-to-end figure is for a batched, GPU-forward-bound run, where preprocessing is 9.5% of the time. Per-image serving and faster GPUs are a different regime, not measured here.

CUDA-event timing (warmup 3, median of 10) on a shared GPU, so throughput is an estimate. Source: results/kernels/*.json.

# Backup: a 90 ms app frame and int8 fidelity

**Median time per stage (ms)**




| CONCH + MUSK student, int8 | Balanced accuracy |
|---|---|
| On the board | **93.05%** |
| x86, fused int8 kernels off | 93.01% |
| x86, onnxruntime defaults | 92.19% |
| fp32 student | 94.15% |

Board vs x86 kernels off: 13 of 7,180 calls differ.

Board flag: 10.2% of tiles, catches 57.2% of its own errors, agreement with the teacher flag 79.6% (κ 0.32).

The device's two readers are two heads on one shared backbone.

Full app loop 25.4 tiles/s. Each Bridge RPC call to the MCU costs about 8 ms. Source: results/uno_q/onboard/*.json.